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Leaching and ponding of viral contaminants following land application of biosolids on sandy-loam soil

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A B S T R A C T

Much of the land available for application of biosolids is cropland near urban areas. Biosolids are often applied on hay or grassland during the growing season or on corn ground before planting or after harvest in the fall. In this study, mesophilic anaerobic digested (MAD) biosolids were applied at 56,000 L/ha on a sandy-loam soil over large containment lysimeters seeded to perennial covers of orchardgrass (*Dactylis glomerata* L.), switchgrass (*Panicum virgatum*), or planted annually to maize (*Zea mays* L.). Portable rainfall simulators were to maintain the lysimeters under a nearly saturated (90%, volumetric basis) conditions. Lysimeter leachate and surface ponded water samples were collected and analyzed for somatic phage, adenoviruses, and anionic (chloride) and microbial (P-22 bacteriophage) tracers. Neither adenovirus nor somatic phage was recovered from the leachate samples. P-22 bacteriophage was found in the leachate of three lysimeters (removal rates ranged from 1.8 to 3.2 $\log_{10}$/m). Although the peak of the anionic tracer breakthrough occurred at a similar pore volume in each lysimeter (around 0.3 pore volume) the peak of P-22 breakthrough varied between lysimeters (<0.1, 0.3 and 0.7 pore volume). The early time to peak breakthrough of anionic and microbial tracers indicated preferential flow paths, presumably from soil cracks, root channels, worm holes or other natural phenomena. The concentration of viral contaminants collected in ponded surface water ranged from 1 to 10% of the initial concentration in the applied biosolids. The die off of somatic phage and P-22 in the surface water was fit to a first order decay model and somatic phage reached background level at about day ten. In conclusion, sandy-loam soils can effectively remove/adsorb the indigenous viruses leached from the land-applied biosolids, but there is a potential of viral pollution from runoff following significant rainfall events when biosolids remain on the soil surface.

1. Introduction

Approximately 5.6 million dry tons of biosolids are generated annually in the U.S. and 60 percent are applied on land (NRC, 2002). Significant levels of pathogens are present in treated biosolids at the time of application (Sidhu and Toze, 2009; Viau and Peccia, 2009; Wong et al., 2010; Viau et al., 2011). Land applied biosolids with a high pathogen load can contaminate surface water by runoff and movement through subsurface drains during rainfall. Adenovirus is a common cause of gastroenteritis, upper and lower respiratory system infections and conjunctivitis (Jiang, 2006), and it was found to be the most prevalent enteric virus in biosolids (Wong et al., 2010). The presence of adenovirus and other enteric viruses in surface water and groundwater has been reported in earlier work (Castignolles et al., 1998; Chapron et al., 2000; Jiang et al., 2001; Borchardt et al., 2003; Xagoraraki et al., 2007).

In the upper Midwestern states of the USA, drainage improvement is a common practice (Fausey et al., 1995). The Great Lakes region include three of the top four states (Illinois, Indiana, Iowa and Ohio) practicing drainage improvement (Pavelis, 1987). Drainage improvement in these states accounts for over 20.6 million hectares of drained land, representing over 37 percent of the total cropland in the region. Several researchers (Ball-Coelho, 2005; Dean and Foran, 1992; Fleming and Bradshaw, 1992; Evans and Owens, 1972) have reported that the application of liquid manure to tile-drained fields resulted in elevated levels of bacteria in the receiving surface waters compared to sites where no manure was applied.

Column experiments have been used in laboratory studies to evaluate the effect of organic matter, water saturation, ionic strength, pH, soil texture and other factors on downward pathogen

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transport in the soil environment (Cheng et al., 2007; Chu et al., 2001; Powelson et al., 1991; Jin et al., 1997, 2000; Zhuang and Jin, 2003; Chetochine et al., 2006). Because laboratory soil columns are screened and packed to a uniform density, they lack the variability of undisturbed soil and may not always be representative of field conditions. For example, even though clay particles are small with great filtering and adsorption capacity, undisturbed clay soil was associated with lower microbial removal than sandy and silt soils in some lysimeter studies (Aislabie et al., 2001; McLeod et al., 2001; Carlander et al., 2000). Carlander et al. (2000) and Pang (2009) reported that the formation of macropores and preferential flow paths from shrinking and cracking of clay soils was the reason for inefficient microbial removal. Pang (2009) compared microbial removal reported in soil column and field studies and reported column experiments had one- to three-log greater microbial removal than field studies.

Several researchers have evaluated lysimeters containing undisturbed soil to investigate manure contaminant and microbial tracer transport in soil systems (Aislabie et al., 2001; Carlander et al., 2000; McLeod et al., 2001, 2003, 2004; Pang et al., 2008; Jiang et al., 2008). Soil depth in these lysimeters ranged from 40 to 100 cm with diameters ranging from 30 to 60 cm (volume ranged from 0.028 to 0.28 m^3). Most studies were performed indoors except for the work reported by Jiang et al. (2008) and Carlander et al. (2000). The focus of their work was on the transport of fecal coliforms, *Escherichia coli* and salmonella phage from dairy manure or pure cultures. To the best of our knowledge, lysimeter studies focusing on the movement of viruses and indicator organisms from biosolid application on cropland under field conditions have not yet been reported.

The specific objectives of this work were to: 1) evaluate the movement of biosolid-indigenous viruses (somatic phage and adenovirus), and anionic (chloride) and microbial (P-22 bacteriophage) tracers through the soil under nearly saturated conditions, and 2) evaluate the viral concentration of ponded surface water after land application of biosolids.

2. Methods and materials

2.1. Lysimeters

Six stainless steel containment lysimeters enclosing a monolith of undisturbed soil were installed in large experimental plots (600 m^2) on a *Kalamazoo fine-loamy, mixed mesic Typic Hapludalfs* soil at the Kellogg Biological Station of Michigan State University (MSU) in southwest Michigan. The lysimeters were used from 1994 to 1999 in a study of nitrate movement to groundwater in a rotation of corn (*Zea mays* L.) and alfalfa (*Medicago sativa* L.) with various treatments of compost, manure or inorganic fertilizer (Basso and Ritchie, 2005). They have not been used for experimental work since that time. From 2000 to 2003, corn was grown continuously on all plots and the lysimeter leachate was evacuated annually. Neither manure nor compost has been applied to the plots since 1999.

The lysimeters are 1.5 m wide and 2.1 m deep. A drainage tube extends from the bottom of each lysimeter to the plot edge to avoid unnecessary disturbance of the cropped area. The bulk density of the soil ranged from 1.5 to 1.6 g/cm^3 (Basso and Ritchie, 2005) and the porosity was approximately 40% on a volumetric basis. The pore volume of each lysimeter was approximately 1.48 m^3 based on a particle density of 2.65 g/cm^3.

2.2. Biosolids

Biosolids from the Plainwell (2008) and St. Clair (2009) wastewater treatment (WWTP) plants (Michigan, US) were applied to the surface (56,100 L/ha) of the lysimeters during the growing season. This application rate was selected as representative of the rate applied by commercial haulers in the region. The biosolids were stored at 4 °C and applied within 24 h of being received from the WWTP. Immediately before the application the biosolids were spiked with the P-22 bacteriophage to a concentration of 3.00×10^{11} and 1.25×10^{10} PFU/100 ml in 2008 and 2009, respectively (Table 1). Subsamples were analyzed within 24 h for concentrations of somatic phage (2009 only); adenovirus, P-22, and percent total solids (Table 1).

2.3. Lysimeter experiments

Six lysimeters were used: lysimeter numbers L1 to L3 in August, 2008 and L4 to L6 in May, 2009. Biosolids were applied on all lysimeters except L3, which was a control. Leachate samples were collected in 2008 and 2009; surface water and core soil samples were collected in 2009. In 2008, biosolids were applied in mid-August within a few days of harvesting the orchardgrass and switchgrass surface biomass. In 2009, biosolids were applied in early May at the beginning of the normal growing season for orchardgrass and switchgrass. Biosolids were applied where corn grain had been grown in 2008 and the corn residue had been tilled after harvest. There was little crop residue on the surface and the 2009 crop had not yet been planted. Biosolids are often applied as a nutrient source and soil amendment prior to planting corn. Leachate and surface samples were monitored for somatic phage (2009 only), adenovirus (2009 only) and P-22 (2008 and 2009).

One mole of anionic tracer (Cl^- as potassium chloride) was mixed in 4 L of water and applied uniformly to the surface of each lysimeter followed by 12.7 mm simulated rainfall on the day before the biosolids were applied. The anionic tracer was only monitored in leachate samples.

The biosolids were applied uniformly (9.9 L/lysimeter) to the soil surface and allowed to remain undisturbed for 12 h prior to simulated rainfall. The uniformity of the application was controlled by using small containers to evenly distribute the biosolids on the lysimeter surface. The rainfall simulator applied water at a rate of 5 cm/h on a semi-continuous (on-off) basis controlled by the operator to minimize surface ponding. A rain gauge was used to monitor the amount of water applied on each lysimeter and the

Table 1
Initial somatic phage, P-22 and adenovirus concentration in biosolids.

Year of study	% Solid	Somatic phage[a]		P-22[b]		Adenovirus[a]	
		PFU per		PFU per		Copies per	
		100 ml	g (dry)	100 ml	g (dry)	100 ml	g (dry)
2008	5.0	NA		3.00×10^{11}	6.00×10^{10}	4.20×10^{8}	8.40×10^{7}
2009	6.0	8.00×10^{4}	1.33×10^{4}	1.25×10^{10}	2.08×10^{9}	3.30×10^{7}	5.50×10^{6}

NA = not available.
[a] Indigenous.
[b] Spiked.

water application rate was approximately 8–10 cm per day. Immediately after biosolids application, irrigation and water sampling continued on a daily basis for about 12 h/day until about 1.7 pore volumes of leachate was collected from each lysimeter. It took approximately 21 days to collect 1.7 pore volume of drainage effluent.

The average ambient temperature during the 2008 and 2009 study was 13.3 °C (ranging from 7.2 to 19.4 °C) and 15.0 °C (ranging from 8.3 to 20 °C), respectively. The average surface soil temperature during the 2008 and 2009 study was 10.0 °C (range from 5.0 to 16.8 °C) and 11.3 °C (ranging from 5.9 to 19.2 °C), respectively. Three sensors (5TM Soil Moisture and Temperature Sensor, Decagon Devices Inc.) were installed in each lysimeter at the depth of 10 cm, 30 cm and 100 cm to monitor soil moisture. The water saturation in each lysimeter soil was about 90% (volumetric basis) during the study period.

Leachate samples (3.9 L) were drawn from the bottom of the lysimeters with a peristaltic pump every 0.1 pore volumes (148 L) and stored in sterilized containers, placed on ice and transported to the laboratory for analysis. The leachate collected between each 0.1 pore volume sample was discharged downslope and at least 5 m from the lysimeters. The leachate volume was recorded in a -graduated 18.9 L container. Approximately 0.1 pore volume of water leached through each lysimeter for every 24 to 36 h. In 2009, leachate samples were also collected once a month for three consecutive months after completion of 1.7 pore volumes sampling.

A circular, galvanized steel ring (137 cm by 30 cm) was placed over the lysimeter on the soil surface of L4, L5 and L6 to retain ponded surface water. Ponded surface water samples were collected daily during the sampling period. Samples (3.9 L) were collected in sterilized containers, stored on ice and transported to the laboratory for analysis each day.

2.4. Infiltration rate

The *in situ* water infiltration rate was measured with a double-ring infiltrometer (ASTM D3385 − 09) upon completion of the experiment and before the soil cores were removed from the lysimeters.

2.5. Chloride analysis

Chloride concentration in the leachate samples was analyzed with an ion selective electrode (model no. 27502-13, Cole Parmer). The electrode was calibrated by plotting millivolts versus three standard chloride solutions on a semi-log scale. The developed calibration curve was used to determine the chloride activity of the leachate samples.

2.6. P-22 propagation and phage analysis

Salmonella phage (P-22) was used as the microbial tracer. The P-22 was propagated by infecting its host strain *S. typhimurium* overnight in Tryptic Soy Broth (TSB; Difco) at 37 °C and isolated by filtering through 0.45 μm cellulose ester-based membrane (Millipore, MA) to remove cell debris. Somatic phage and P-22 were analyzed by the double layer agar method (USEPA, 2001). The host cell for somatic phage was *E. coli* CN13. All dilutions were made with sterilized phosphate buffer water (PBW).

2.7. Adenovirus analysis

Water samples were concentrated to enhance adenovirus detection during the qPCR reaction. The concentration method developed by Haramoto et al. (2005) was used with the following modification: Amicon Ultra (Millipore, Billerica MA) rather than Centriprep YM-50 was used to concentrate the NaOH eluent. The filtered volume for surface and groundwater was 100 ml and 2 L, respectively. The final volume of concentrated eluent was around 140 μL and it was stored at −80 °C for DNA extraction. The primers and probe were adopted from Heim et al. (2003). Each qPCR reaction mix included 10 μL of 2× LightCycler 480 TaqMan Master Mix; 1.0 μL of each forward and reverse primer (each final concentration was 500 nM); 0.6 μL of 10 μM TaqMan probe (final conc. = 300 nM); 2.7 μL of PCR-grade water and 5 μL of DNA sample or standard. The real-time PCR running program (all thermocycles were performed at a temperature transition rate of 20 °C/s) was 95 °C for 15 min followed by 45 cycles at 95 °C for 3 s; 55 °C for 10 s; 60 °C for 60 s and a final cycle of 40 °C for 30 s. The fluorescent signal was detected after each extension cycle. The efficiency, slope and *R* square of the qPCR standard curve were 93.4%, −3.49, and 0.9992, respectively. The detection limit was 10 copies per reaction. Positive and negative controls were included in each reaction run. The qPCR standards were constructed by cloning the target sequence into the plasmids and the detailed procedure was described in Wong et al. (2010).

2.8. Viral analysis of soil samples

Intact soil cores (3.8 cm diameter) were extracted to a depth of about 90 cm after completion of the rainfall simulation experiments in 2009 (volume of soil cores ranged from 6.9 to 10.3×10^2 cm^3). The soil probe was only able to extract the soils above the 90 cm depth since the soil below that depth was primarily unconsolidated sand and the probe was not able to retain an intact core. The soil cores were subdivided based on identifiable changes in soil color or texture. Soil physical and chemical properties and residual virus concentrations (L4, L5 and L6) were analyzed. No virus analyses were performed on L1, L2 and L3 soil samples because the core soil samples were taken more than one year after the completion of the 2008 experiments. Analysis of the soil physical and chemical properties (Table 2) was done by the Soil and Plant Nutrient Laboratory at MSU.

Viruses were eluted from the soil samples by stirring 50 g of soil in 50 ml of 10% beef extract for 30 min (Williamson et al., 2003). This method has been used to recover viruses from soils in previous studies (Jin et al., 1997; Zhuang and Jin, 2008). The solid phase of the mixture was spun down by centrifugation at 10,000 × g for 30 min at 4 °C and the supernatant was retained for virus analysis.

2.9. Recovery of the Cl tracer and removal rate of P-22

Mass recovery of the anionic and microbial tracers was calculated by the trapezoidal rule whereby the area under the curve of the effluent concentration (ppm of Cl^- or virus/L) versus pore volume (0.1 pore volume = 148 L) was measured and normalized with the initial mass input to the system.

The removal rate of P-22 was calculated using Equation (1) which describes the relative log-reduction in microbial concentration per unit of distance traveled (Pang, 2009).

$$\lambda = \frac{-\ln\,(\text{Mass recovery})}{x} \tag{1}$$

Where:

λ is the removal rate ($\log_{10}$/m)
x is the soil depth (m)
Mass recovery is (dimensionless)

Table 2
Physical and chemical characteristics for each lysimeter soil applied with biosolids.

Soil depth (cm)	pH[a]	CEC[b] (meq/100 g)	OM[c] (%)	Sand (%)	Silt (%)	Clay (%)	Soil classification	P[d] (ppm)	K[b] (ppm)	Mg[b] (ppm)	Ca[b] (ppm)
L1											
0–25	6.4	7.7	1.8	55.2	35	9.8	Sandy loam	54	131	165	1196
25–46	7.1	7.7	1.4	53.2	35	11.8	Sandy loam	54	69	157	1235
46–61	7.3	9.1	1.1	56.4	28.8	14.8	Sandy loam	52	95	180	1474
61–91	7.1	8.7	0.6	71.8	9.4	19.8	Sandy loam	28	18.1	79.1	1372
L2											
0–25	6.9	8.2	1.1	62.4	23.8	13.8	Sandy loam	28	95	196	1273
25–43	7.1	8.8	1.2	54.8	29.4	15.8	Sandy loam	36	83	210	1368
43–76	6.9	11	0.9	67.8	9.4	22.8	Sandy clay loam	27	120	238	1735
L3											
0–15	7.0	6.4	1.8	53.2	31.8	15.0	Sandy loam	53	107	217	865
15–36	7.0	7.9	1.3	43.4	33.8	22.8	Loam	35	74	251	1112
36–51	6.9	9.6	1.1	57.4	13.8	28.8	Sandy clay loam	31	92	312	1355
51–61	5.9	6.4	0.6	79.9	5.7	14.4	Sandy loam	43	68	177	706
L4											
0–23	7.2	9.1	2.1	52.4	36.8	10.8	Sandy loam	66	111	218	1408
23–41	7.2	11.6	1.1	52.8	25.4	21.8	Sandy clay loam	28	115	249	1843
41–51	7.2	10.5	0.9	68.8	12.8	18.4	Sandy loam	29	106	228	1971
51–76	7.2	3.5	0.5	86.9	3.2	9.9	Loamy sand	35	54	98	519
L5											
0–20	6.8	8	2.2	53.8	35.8	10.4	Sandy loam	64	126	201	1210
20–36	7	9	1.2	43.8	32.8	23.4	Loam	40	76	216	1393
36–51	6.9	11.7	1.1	47.8	25.4	26.8	Sandy clay loam	29	127	303	1769
51–66	6.9	8.6	0.8	73.9	7.7	18.4	Sandy loam	30	77	166	1399
66–84	7	4.5	0.5	88.1	2.5	9.4	Loamy sand	37	38	92	733
L6											
0–15	6.6	7	2	51.2	36	12.8	Sandy loam	50	104	171	1062
15–25	6.9	6.6	1.3	51.4	33.8	14.8	Sandy loam	56	57	162	1016
25–46	7.1	9.3	1.2	52.4	32.8	14.8	Sandy loam	64	90	242	1415
46–61	7	7.7	0.8	54.8	21.4	23.8	Sandy clay loam	41	67	200	1164
61–74	7.1	8	0.6	69.8	11.4	18.8	Sandy loam	40	75	210	1205
74–91	7.1	7.5	0.6	63.8	16.4	19.8	Sandy loam	39	68	188	1155

CEC = Cation exchange capacity; OM = Organic matter; P = Phosphorus; K = Potassium; Mg = Magnesium; Ca = Calcium.
[a] 1:1 soil:water ratio.
[b] Ammonium acetate method (Thomas, 1982).
[c] Dichromate method (Nelson and Sommers, 1982).
[d] Bray P1 method (Frank et al., 1998).

2.10. Recovery of P-22 from soil

The recovery of P-22 from the soil cores (depth of 76–91 cm) was calculated as the sum of P-22 recovered from each soil layer normalized with the initial mass of P-22 applied to the soil surface. The P-22 recovered in each layer of soil was calculated as:

$$M_{\text{soil}} = \frac{(A \times D \times (1-\theta) \times B \times C_{\text{soil}})}{0.26} \tag{2}$$

Where:

M_{soil} is the P-22 recovered in each soil layer (PFU)
A is the surface area of the lysimeter (cm^2)
D is the thickness of each layer of core soil samples (cm)
θ is the soil porosity ($V_{\text{pore}}/V_{\text{total}}$) = 0.40
B is the bulk density (1.6 g/cm^3)
C_{soil} is the P-22 concentration in each layer (PFU/g)
0.26 is the recovery of virus from soils with beef extract (Williamson et al., 2003).

Equation (2) assumes a homogenous distribution of P-22 in each soil layer. Porosity and soil bulk density values are based on Basso and Ritchie (2005).

2.11. Decay analysis

The P-22 concentration in the ponded surface water was fit to a first order decay model [Equation (3)]:

$$\ln(C_t) = -Kt + \ln(C_o) \tag{3}$$

Where:

C_t is the microorganism concentration (pfu/100 ml)
t is time from application (days)
C_o is the phage concentration (pfu/100 ml) at time zero
K is the decay coefficient (d^{-1}).

3. Results

3.1. Soil properties

The physical and chemical properties of the soil are listed in Table 2. The soil series was a Kalamazoo Loam (*Kalamazoo fine-loamy, mixed mesic Typic Hapludalfs*). The near-surface soil texture was typically a sandy-loam although most of the lysimeters had a sandy-clay-loam layer at depths ranging from 20 to 40 cm. The sand content generally increased at greater depth.

The Natural Resource Conservation Service (USDA-NRCS, 2012) drainage classification for the soil series was 'poor'. Measured infiltration rates ranged from 3.6 to 8 mm/h for the lysimeters (L1–L5) containing orchardgrass or switchgrass (Table 3). The measured infiltration rate for L6 (corn) was unrealistically low (1.0 mm/h) and was not consistent with the flow rate through the lysimeter during the water sampling period. There was no crop residue, no vegetative cover and no active root system to protect the soil from the impact of the simulated rainfall. After the soil

Table 3
Infiltration rates, drainage classification, root system of each lysimeter.

Lysimeter	Infiltration rate (mm/h)	Drainage class	Crop
L1	5.9	Poor	Orchardgrass
L2	3.6	Poor	Orchardgrass
L3 (control)	8.0	Poor	Orchardgrass
L4	7.7	Poor	Switchgrass
L5	4.3	Poor	Switchgrass
L6	1.0	Poor	Corn

Control had no biosolids application.

dried, the resulting soil compaction and loss of pore space restricted the infiltration rate.

3.2. Lysimeter effluent

The breakthrough curve (BTC) of the chloride tracer revealed differences in the flow characteristics of each lysimeter (Fig. 1). This variability was expected given the variation in soil texture and vegetation. The peak concentration occurred near 0.3 pore volume (PV) for all lysimeters and remained above background levels throughout the sampling period. Chloride recovery ranged from about 33 to 99% (Table 4).

Neither adenovirus nor somatic phage was recovered from the leachate samples. P-22 was recovered from L2 (orchardgrass), L5 (switchgrass) and L6 (continuous corn) leachate, but none was detected in L1 (orchardgrass), L4 (switchgrass) or L3 (control) (Table 4). The P-22 breakthrough curves of L2, L5 and L6 are shown in Fig. 2. The peak breakthrough in L2, L5 and L6 occurred at 0.7, 0.3 and <0.1 PV, respectively. The rapid breakthrough in L6 may indicate the potential for preferential flow through transient flow paths when biosolids are applied to an annual crop such as maize whereby the root system decays leaving channels for preferential flow. There was a three- to four-log reduction of P-22 from the initial concentration in the spiked biosolids to the peak concentration in the leachate. The recovery of P-22 in L2, L5 and L6 leachate was 0.59, 0.12 and 2.14% of the initial concentration (Table 4), respectively. The P-22 removal rate (λ) for L2, L5 and L6 was 2.4, 3.2 and 1.8 log/m, respectively.

3.3. Viral levels in soil samples

The soil cores extracted from (L4, L5 and L6) were evaluated for P-22, somatic phage and adenovirus. Neither somatic phage nor adenovirus was detected, but P-22 was present in every sample. Interestingly, no P-22 was found in L4 leachate, but the concentrations of P-22 extracted from the L4 soil samples were greater than L5 and L6 at nearly every depth (Fig. 3).

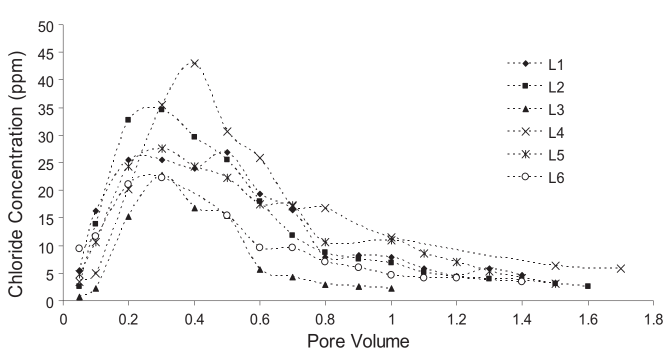

Fig. 1. Breakthrough curves for the chloride tracer.

Table 4
Recovery percentage of chloride and P-22 from leachate and soils above the 90 cm depth.

lysimeter	Leachate		Soil[a]
	Chloride	P-22	P-22
L1	71.3	UD	ND
L2	74.2	0.59	ND
L3	32.6	UD	ND
L4	99.3	UD	0.19
L5	74.2	0.12	0.011
L6	51.2	2.14	0.012

UD = undetected; ND = not done.
[a] No chloride analysis on soil samples.

3.4. Ponded surface water

The somatic phage and P-22 concentrations in the ponded surface water are shown in Fig. 4. Somatic phage reached non-detectable levels around day ten. Adenoviruses were detected in all surface water samples but no decay trend was observed. The average concentration of adenovirus in the surface water from all sampling events was $2.94 \pm 3.05 \times 10^3$ copies/100 ml, about 4 logs lower than the initial concentration in biosolids. It took longer for

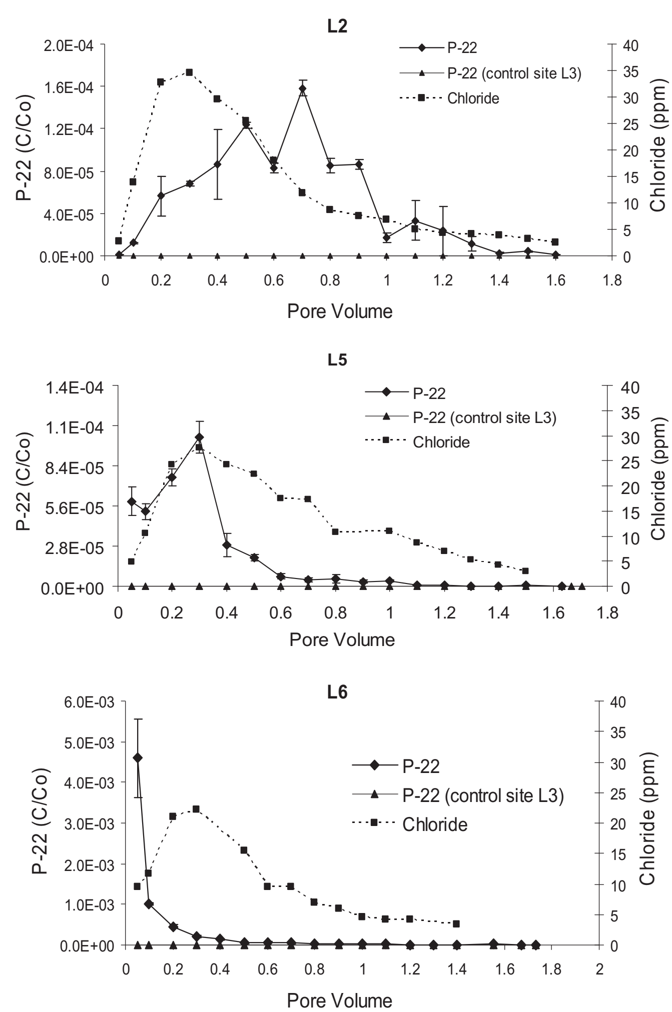

Fig. 2. Breakthrough curves of P-22 and the chloride tracer in L2, L5 and L6 leachate. Error bars represent the standard deviation of the duplicate measurements from each sample. C/C_0 = P-22 concentration in the effluent/initial P-22 concentration in the biosolids.

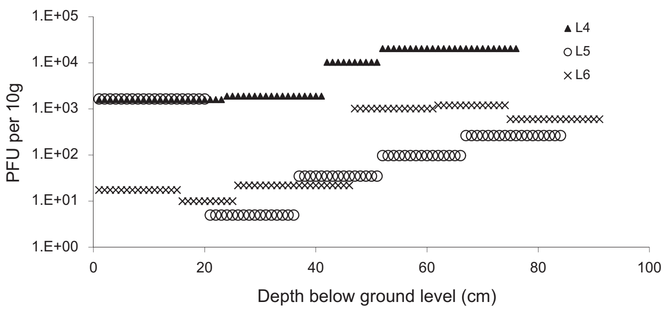

Fig. 3. P-22 concentration at different soil depths in recovered soil cores: no somatic phage or adenovirus was detected.

P-22 to reach non-detectable levels (>21 days) (Fig 4), likely because its initial concentration was 6 logs greater than the concentration of somatic phage (Table 1).

The greatest viral concentration in the ponded water was at the beginning of the simulated rainfall and the samples with the greatest concentration were about 1–10% of the initial concentration in the spiked biosolids. The reduction of viruses in the ponded water samples over time was fit to a first order decay model (Fig. 5) and the resulting decay coefficients of somatic phage and P-22 were similar (slightly less than 0.40/day).

4. Discussion

To the best of our knowledge, the leaching of biosolid-associated pathogens through undisturbed soil in large-scale lysimeters with soil depths greater than 100 cm has not been reported. The large-scale lysimeter allowed a unique opportunity to collect data on

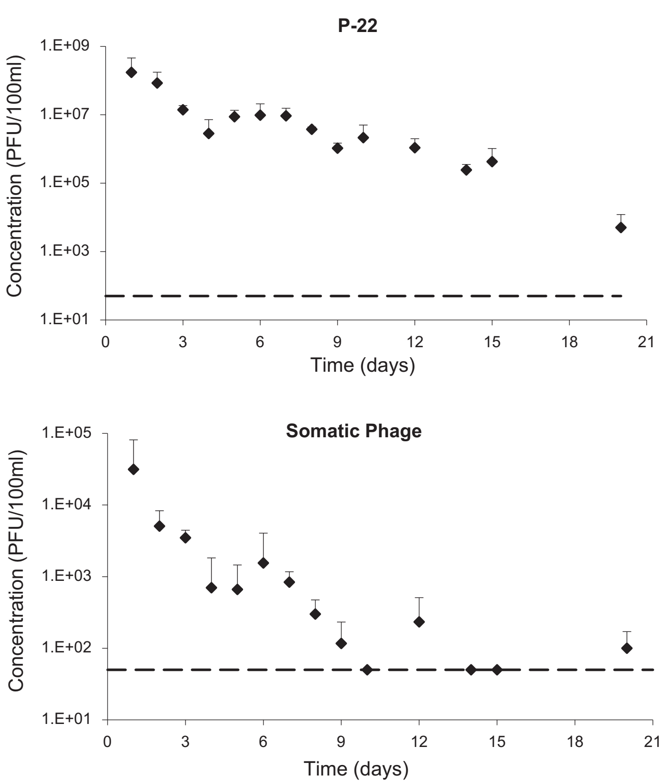

Fig. 4. P-22 and somatic phage levels in ponded surface water samples in 2009. The dotted-line indicates the detection limit. Error bars indicate one standard deviation from the mean of L4, L5 and L6.

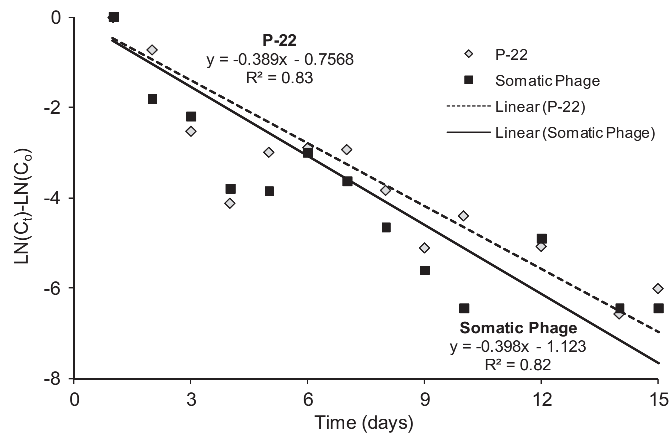

Fig. 5. Decay curves of P-22 and somatic phage in ponded surface water samples from L4, L5, and L6.

the movement of microbial contaminants under field conditions. Because only six lysimeters were available for land application of biosolids, multiple crops were not replicated. It is not uncommon to have large-scale field lysimeter data that is not replicated but with a qualitative interpretation of their results (Basso and Ritchie, 2005; Rasse et al., 2000; Martin et al., 1994). The results obtained from the freely drained lysimeters in this study can be particularly applicable to artificially drained cropland.

The rapid, peak breakthrough of the anionic and microbial tracers (<1 PV) indicated transport through preferential flow paths. Preferential flow was reported in earlier lysimeter work (Aislabie et al., 2001; McLeod et al., 2001, 2003, 2004; Pang et al., 2008; Jiang et al., 2008). The transport rate measured in laboratory scale experiments may be considerably lower than the transport rate in the natural environment because the preferential flow pathways allow contaminants to bypass the soil matrix. Besides natural soil cracks and fissures, the root systems of vegetative covers also improve infiltration by providing root channels for preferential flow.

The difference in the BTC and recovery between the anionic and microbial tracers was largely due to differences in sorption behavior. Even though viruses are small (usually less than 100 nm), they can aggregate under the conditions that favor neutralization of their surface charge such as an aqueous pH near the isoelectric point of the virus and the present of multivalent cations (Wong et al., 2012; Gutierrez et al., 2010; da Silva et al., 2011). Large virus aggregates may be more readily retained by subsurface soil compared to individual viruses or an anionic tracer.

The removal rates of the P-22 bacteriophage in three of the lysimeters (1.83–3.21 log/m) were similar to those reported by Jiang et al. (2008; 1.92 to 2.80 log/m) and Carlander et al. (2000; 3.76 log/m). Manure-associated fecal coliforms were detected in the leachate samples reported by Jiang et al. (2008) but neither biosolids-associated somatic phage nor adenovirus was detected in the leachate from our lysimeters, likely because of the greater depth of soil (2.1 m versus 0.4–1.0 m). The results of our work with the application of biosolids on a sandy loam soil confirm the results of Jiang et al. (2008) and Carlander et al. (2000) with dairy manure on similar soils. This indicates that a sandy loam soil with a vegetative cover can be an effective filter for removing enteric viruses, but depth of the soil profile is important.

In addition, the rainfall rate (8–10 cm/day) simulated in this study was much higher than the natural rainfall rate in order to maintain a nearly saturated condition in the lysimeters; therefore, the removal of viral pathogens in similar soils during natural rainfall events would be greater because unsaturated soils are known to have higher microbial removals (Jin et al., 2000; Chu et al., 2003).

The low recovery of P-22 (Table 4) and no recovery of indigenous viruses from the leachate and soil cores indicate that most viruses were sorbed to sludge particles and eventually decayed. Biosolids spiked with adenovirus were applied on lysimeters in a previous study, and results showed that low concentrations of adenovirus were found in the leachate samples (Horswell et al., 2010). The authors speculated the results were due to the irreversible sorption of viruses to sludge particles. In addition, Chetochine et al. (2006) reported that most of the indigenous phages remained in the solid pellet after a series of extractions, and Gerba et al. (1980) reported that enteroviruses formed strong attachments to sludge particles and were difficult to elute. Sano et al. (2004) reported that the virus-binding proteins (VBPs) in a bacterial culture from activated sludge played a key role in attaching indigenous viruses to sludge particles. Therefore, indigenous viruses, adenovirus and somatic phage, could form a stronger attachment to the sludge solids than the spiked P-22 which resulted in lower partitioning to the water phase. Another possible reason for no detection of adenovirus was the low recovery of the sampling process. According to Fong et al. (2010), the recovery of adenovirus in MilliQ water and a river water matrix ranged from 0.17 to 6.98 percent using HA filtration, much lower than the 30 to 74 percent recovery reported earlier (Haramoto et al., 2005). Overall, our results show that it is important for future studies to pay more effort to understand the transport and retention of indigenous viruses as opposed to solely using spiked viral indicators to determine the virus transport behaviors in the natural environment.

Because the impact of rainfall breaks down soil particles there was mixing of water, soil and biosolids at the soil surface during the simulated rainfall. The ponded water samples included viruses attached to soil particles, sludge particles, and unattached viruses or aggregates. Each of these are sources of contaminants that can be transported in overland flow (Tyrrel and Quinton, 2003; Muirhead et al., 2005). Based on the our results, the virus concentration in ponded surface water can be as great as 1–10% of the initial virus concentration in the biosolids. These virus concentrations represent a considerable threat to water quality from surface runoff if biosolids are allowed to remain on the soil surface after application. The biosolid-associated pathogens may exist for several days under wet conditions. In earlier work (Pourcher et al., 2007), no enteroviruses in sludge contaminated soil were detected beyond 14 days after application, but adenovirus was detected after 20 days in our study. The greater survival in our work may have been due to the strong UV resistance of adenoviruses and the presence of vegetation, crop residue or soil particles that may have created a protective micro-environment.

5. Conclusions

This study showed that a sandy-loam soil with a depth of 2 m or greater can effectively remove/adsorb the indigenous viruses leached from land applied biosolids. The early breakthrough of P-22 indicated preferential flow pathways are important in the transport of the viruses in the soil. The movement of anionic tracers was not a reliable predictor of the movement of viruses, likely because of differences in sorption behavior and physical size. Finally, somatic phage, adenovirus and P-22 were detected in ponded surface water for at least two weeks; therefore, runoff from cropland where biosolids were not well incorporated soon after application may be a threat to surface water quality.

The results presented in this paper can contribute to the development of best management practices for the land application of manure and biosolids on artificially drained land. Some of the management practices that may help protect water quality are: 1) pre-tillage to disrupt the continuity of macro-pores, 2) controlled (low) application rates, 3) incorporation of biosolids during land application, and 4) timing biosolids/manure application rates to avoid application on wet ground, when tiles drains are flowing, or when there is a chance of significant rainfall (>10 mm) within the next few days.

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